VolAnti uses sound analysis to identify drones

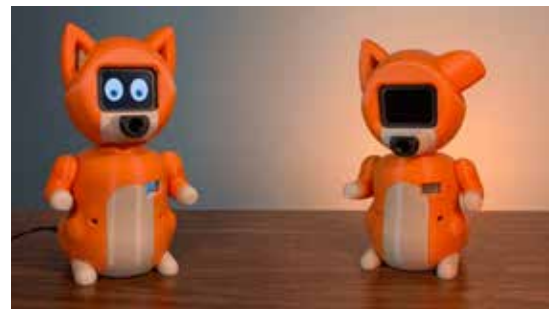
VolAnti is an open source drone detection project by Agam Rossen that uses sound analysis to identify drones without relying on their radio transmissions. Its publicly available resources include firmware, printed circuit board (PCB) manufacturing files, a bill of materials, enclosure designs, test audio, and a browser-based simulator. This approach also makes the project relevant to detecting fibre-optic FPV drones, which can evade conventional radio-frequency detection because their control signals travel through a physical fibre-optic link. The device only detects and reports potential drones; it does not jam signals or attempt to intercept them.

The project's hardware is built around an ESP32-S3 microcontroller and four ICS-43434 digital MEMS microphones arranged in a cross. Their audio streams are synchronised and combined to improve the signal-to-noise ratio before analysis. The project includes a LoRa radio module for sharing alerts between nearby units, an e-paper display, a buzzer and vibration alerts. Power comes from a rechargeable 2,500mAh lithium-polymer battery, with approximately 18–22 hours of operation.

By publishing its firmware, design files, test resources and assembly instructions, VolAnti gives developers a foundation for reproducing the detector, investigating acoustic signal processing and adapting the implementation to different environments.



The robot is designed to keep its core functions on the UNO Q rather than depending on a connected PC. Its Linux-capable side runs the AI pipeline, voice engine and application logic, while the real-time microcontroller handles the physical hardware. Nuvi uses Gemini to understand speech and generate responses, while Piper provides voice output directly on the board. The robot can also search the web when required and interact with smart home devices through voice commands or hand gestures. An open-palm gesture can additionally trigger Nuvi to report the room's temperature and humidity.



The complete project is available through Arduino Project Hub, including 3D files, electronics information, code and a step-by-step build guide. The design can also be customised to resemble a different pet.

Seeed develops six-axis robotic arm

Seeed Studio's reBot-DevArm is an open source six-axis robotic arm designed as a desktop platform for robotics development and experimentation. The arm has six revolute joints plus a seventh motor for its parallel gripper, with the repository providing mechanical CAD files, software, documentation and a bill of materials for building the system.



The project is available in two hardware variants: the B601-DM and B601-RS. The DM version uses Damiao actuators with a 24V power system, reaches up to 767mm and has a recommended payload of 1.5kg. The RS version uses RobStride actuators, operates from 48V, reaches 754mm and has a recommended payload of 2.5kg.

A major part of the project's open source approach is the amount of hardware information released. The repository includes STEP/STP CAD files, STL meshes, URDF descriptions, assembly documentation and component-level BOM information. The arm is also integrated with several robotics software frameworks. Its ecosystem includes ROS and ROS 2, MoveIt, Pinocchio, MuJoCo and NVIDIA Isaac Sim, while the hardware can be used with Hugging Face LeRobot and GR00T N1.7 for robot-learning workflows.

The hardware is licensed under CERN-OHL-W-2.0, while the software uses Apache-2.0.

Cactus Compute's Needle 2 LLM can run locally

Needle 2, developed by Cactus Compute, is a compact, function-calling large language model designed specifically to translate plain-English requests into structured actions. On a Raspberry Pi 5, the 14MB model runs entirely on the CPU, allowing commands such as turning on an LED to be processed locally without a dedicated AI accelerator or cloud API. Raspberry Pi reports a 78ms response time for a simple LED command in its demonstration.